Ieva Bagdonaviciute

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EDUCATION

Stanford University, School of Engineering Expected June 2028
Master of Science in Computer Science, Artificial Intelligence Specialization
University of Illinois at Urbana – Champaign, Grainger College of Engineering May 2026
Bachelor of Science in Computer Engineering, Highest Honors | Undergraduate Researcher Certificate GPA: 3.90/4.0

RESEARCH EXPERIENCE

Stanford University | NeuroAILab Sep 2026 – present
Research Assistant Stanford, CA, USA

- Leading **VLA evaluation** for pick-and-place tasks across physically diverse objects on bimanual YAM robots, establishing a **standardized evaluation framework**.
- Developed a **general-purpose robotics interface** for scalable data collection and experiment control

University of Illinois at Urbana-Champaign | Computer Vision Group Aug 2025 – Aug 2026
Student Researcher Urbana, IL, USA

- Fine-tuned **vision-language models with LoRA** for physical reasoning, investigating dataset scaling, difficulty-aware sample weighting, multimodal embeddings, and prompting strategies to improve model performance.
- Built an end-to-end **RGB-D data and generative reconstruction pipeline**, integrating monocular depth estimation, physics-based simulation, generative models, and automated VLM feedback across multiple datasets and conditions.
- Designed quantitative experiments using **PSNR, SSIM, LPIPS, CIEDE2000, and mixed-effects statistical models** to evaluate model performance and analyze reconstruction quality; contributed to a preprint

Microsoft Research | AI Frontiers Laboratory May 2025 – Aug 2025
Research Intern Redmond, WA, USA

- Built an **end-to-end VLM evaluation pipeline** in Microsoft's open-source EUREKA framework, **preprocessing 10,000+ video samples for large-scale evaluation** and benchmarking **6 frontier models** on physical reasoning.
- Implemented an **LLM-as-a-Judge** system and 6 custom test types to evaluate perception, reasoning, and model failure modes across 3 physical reasoning datasets.
- Developed confidence–accuracy and perception–reasoning analyses with automated evaluation and visualization; co-authored a preprint and presented at the Microsoft Research Intern Poster Session.

University of Illinois Urbana-Champaign | Prof. Yoram Bresler's Laboratory Aug 2024 – May 2025
Research Assistant Urbana, IL, USA

- Developed **Python pipelines for 3D ultrasound/EIT simulation and U-Net medical-image segmentation**, integrating FOCUS, k-Wave, and AbdomenAtlas.
- Generated **20,000+ multimodal parameter maps** from **19 segmented abdominal organs and body-composition tissues** for future multimodal model training.

The Swiss Federal Institute of Technology (EPFL) | Computational Quantum Science Laboratory May 2024 – Aug 2024
Machine Learning Research Intern Lausanne, Switzerland

- **Developed a differentiable MCMC framework in JAX**, implementing Metropolis-Hastings sampling, gradient estimation, and optimization to model the 2D Ising phase transition.
- **Benchmarked 2 gradient-estimation methods** across increasing system sizes, achieving **>96% phase-transition prediction accuracy** on larger systems and characterizing scalability tradeoffs.
- Presented a research report and a poster at the Engineering Research Fair and International Research Symposium

PUBLICATIONS & POSTERS

- *Aghajanzadeh, S., Wang, Y.,* ***Bagdonaviciute, I.****, & Forsyth, D.* (2026). *VLMs Win: A Systematic Evaluation of Underwater Image Reconstruction.* Preprint, arXiv:2608.11425v1 [cs.CV]. https://arxiv.org/abs/2608.11425
- ***Bagdonaviciute, I.*** *& Vibhav Vineet* (2025). *Physics Knowledge in Frontier Models: A Diagnostic Study of Failure Modes.* Preprint arXiv:2510.06251v2 [cs.CV]. https://arxiv.org/abs/2510.06251
- ***Bagdonaviciute, I.*** *& Vibhav Vineet* (2025). *Does Physics Knowledge Emerge in Frontier Models?* Poster presented at the Microsoft Research Intern Poster Session, Redmond, WA.
- ***Bagdonaviciute, I.*** (2024). *Optimization of the Ising Model: Comparing Gradient Estimation Methods.* Poster presented at the Engineering Research Fair and International Research Symposium, University of Illinois Urbana-Champaign.
- ***Bagdonaviciute, I.*** *& Gavin N. R.* (2024). *Analyzing Industrial Software via Open-Source Frameworks*. Poster presented at the Undergraduate Research Symposium, University of Illinois Urbana-Champaign.

INDUSTRY & TECHNICAL EXPERIENCE

University of Illinois at Urbana-Champaign | ECE Department Sept 2023 – May 2024
ECE120 (Introduction to Computing) Course Grader Urbana, IL, USA

- Graded 15+ students' analytical homework on a weekly basis in areas including but not limited to C programming, combinational and sequential logic analysis and design, finite state machines, and the von Neumann model
- Provided constructive feedback to students to foster their critical thinking and academic integrity
- Collaborated with course instructors to ensure fair grading, maintained accurate records, and met strict deadlines to support smooth course operations for 100+ students

Teltonika IoT Group May 2023 – Jul 2023
Embedded Software Engineering Intern Vilnius, Lithuania

- Operated with STM32CubeIDE software to configure a functional STM32F413 microcontroller on the FMB640 hardware platform, adhering to Telematics coding style
- Developed software layers using CMSIS-RTOS API v2 and STM32 low layer library, encompassing drivers (GPIO and UART), APIs (UART, command, debug, LED, and heap), and Application (CLI) tailored for the tracker's optimal performance in reading messages and performing the given commands
- Conducted and delivered weekly research presentations on topics such as UART, RS-485, MQTT, I2C, I2S, CRC, Ethernet and more, showcasing strong analytical and communication skills

AI4ALL – College Pathways Feb 2023 – May 2023
Participant Urbana, IL, USA

- Learned core AI concepts including types of machine learning, gradient descent, and neural networks; examined ethical considerations in data processing and AI implementation
- Built and presented a final project – a visual model for detecting incorrect exercise form from body-movement data

AWARDS & HONORS

- 1st Place Winner, Stanford University Flower.ai Hackathon (2026) – awarded first place for developing a privacy-preserving agentic AI system.
- Undergraduate Researcher Certificate (2026) – awarded in recognition of sustained and significant undergraduate research contributions
- Illinois Engineering Achievement Scholar (2024, 2025) – awarded to top engineering students at UIUC
- Illini Moms Association Scholar (2025) – awarded to 7 UIUC students for academic excellence, leadership, and service
- Excellence Research Scholar (2024) – awarded during research internship at EPFL, Switzerland
- Global Research Scholar (2024) – selected for academic excellence and international research experience
- Phi Eta Sigma Honor Society (2023) – recognized for academic excellence

LEADERSHIP & COMMUNITY INVOLVEMENT

Google Developers Student Club at UIUC Aug 2023 – May 2024
Founder and Lead Urbana, IL, USA

- Collaborated with Google to organize various social events focusing on technology and career development
- Supervised three committees, including software development teams building student-focused solutions such as the carpooling app UNIride and volunteering platform DeedMatch
- Led and built a community of 35+ like-minded and technology-driven people and promoted inclusivity and collaboration by creating an open environment for students from diverse technical and cultural backgrounds

Women in Electrical and Computer Engineering Sep 2022 – May 2026
Technical Committee Member Urbana, IL, USA

- *Engineering Open House 2024 – "Flower Power"*: Led a sub-team of 4 to develop a solar-tracking robot with photoresistors for light sensing, servo-motor panel orientation, and a mobile app override interface; project earned 3rd place in Chevron Environmental and Sustainability Efforts and 2nd place in Most Industry Impact categories
- *Engineering Open House 2023 – "Robot S.U.N.D.A.Y."*: Collaborated on building a six-DOF robotic arm with a Flutter-based mobile app, integrating Bluetooth (HC-05) and Arduino UNO to drive six servo motors

Phi Eta Sigma Honor Society Sep 2023 – May 2026
Scholastic and Professional Development Committee Member Urbana, IL, USA

- Tutored students in ECE 120 and ECE 220 courses, provided mentoring and academic hours, and facilitated professional development events, networking sessions, resume reviews, and skill-building workshops
- Organizing philanthropy and outreach initiatives as well as participating in community impact events